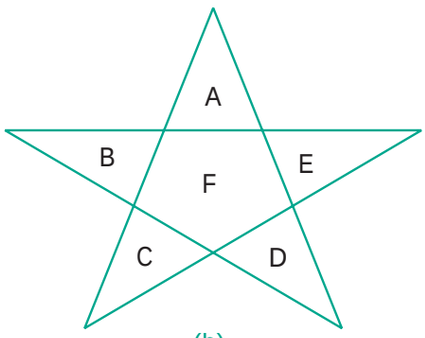
(ii) Solution

A, B, C, D, & E are five triangles.

No combination of these triangles taken 2 at a time, form any triangle.

Only the combinations of A, F & C; A, F & D; B, F & E; B, F & D and C, F & E form triangles when taken 3 triangles at a time. Hence, the total number of triangles = 5 + 5 = 10.



(b)

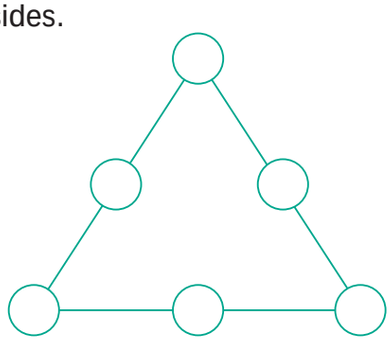
Exercise 6.1

1. Suppose, you have two shorts, one is black and the other one is blue; three shirts which are in white, blue and red. You again wish to make different combinations, but you always want to make sure that the shorts and shirt that you wear are of different colours. List and check how many combinations are possible now.

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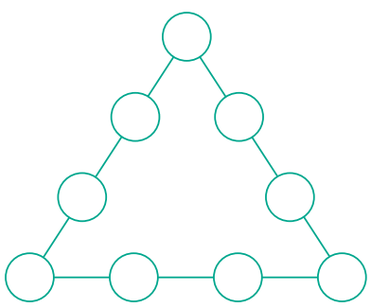
2. You have two red and two blue blocks. How many different towers can you build that are four blocks high using these blocks? List all the possibilities.

3. In the following magic triangle, arrange the numbers from 1 to 6, so that you get the same sum on all its sides.

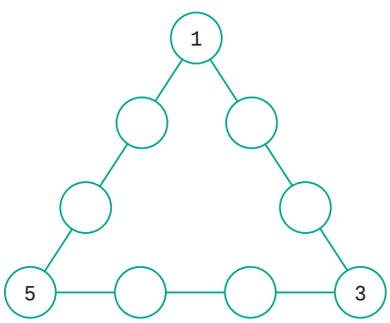


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4. Using the numbers from 1 to 9
   (i) Can you form a magic triangle?
   (ii) How many magic triangles can be formed?
   (iii) What are the sums of the sides of the magic triangle?



5. Arrange the odd numbers from 1 to 17 without repetition to get a sum of 30 on each side of the magic triangle.



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